Galaxy Quenching Follows a Simple Step Function

Gas supply versus gas depletion timescales explain when galaxies stop and restart forming stars.

arXiv:2610.20505 — Step-like quenching from gas mass conservation in 350,000 galaxies

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Background

Galaxies quench when their cold gas runs out, but the many proposed mechanisms—black holes, halos, morphology—have resisted a single unifying description.

No simple framework links diverse quenching processes to one observable response across the galaxy population.

What this paper is selling

Deriving a step-function quenching response from gas mass conservation lets the authors infer gas-supply timescales directly from observed quenched fractions. Black hole growth and halo mass then map cleanly onto that single timescale competition.

Key results to remember

- Characteristic gas-supply timescale is 1 Gyr across ~350,000 low-redshift galaxies.
- Black holes drive inside-out quenching; dark matter halos drive outside-in quenching.
- Intermediate surface-density galaxies sit in a critical state for quenching and rejuvenation.

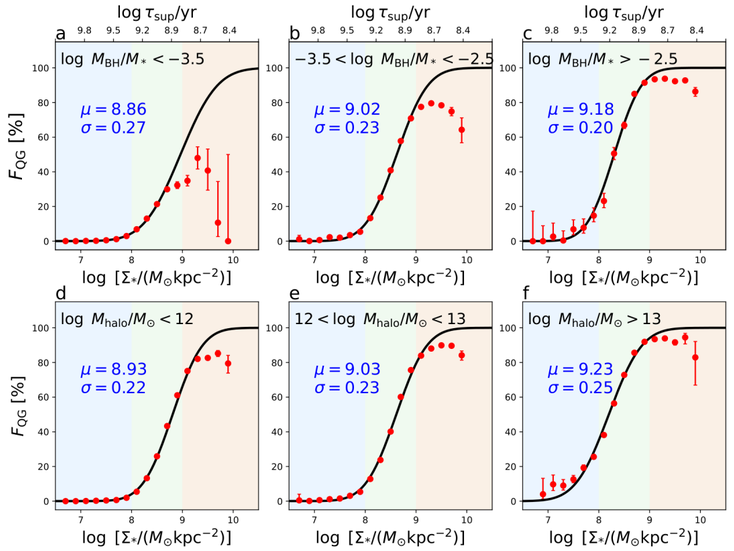


Figure 2 | Black hole growth lengthens and narrows supply timescales, while halo mass lengthens and broadens them, separating their quenching signatures.

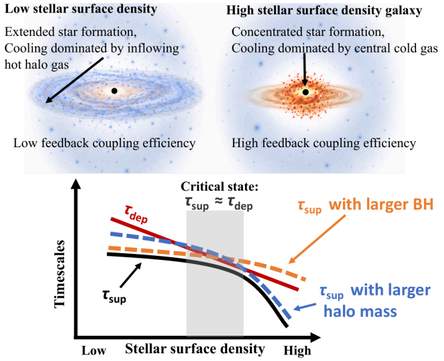


Figure 4 | At intermediate surface density, supply and depletion timescales converge, making galaxies most sensitive to quenching and rejuvenation.